MC

UNIVERSITY

Maya Chen: university portfolio

IB Diploma · Year 13

My map has six lines, but one question runs through all of them: what can careful measurement tell us about the place we live? This page follows how that question grew from a Year 10 science fair into a plan to study Data Science.

AT A GLANCE

Programme	Lines followed	Teacher endorsements	Reflections
IB Diploma	**Data Science, Science, Computing, Environment, Service**	**10**	**25**

Also on my map: Swimming.

Data Science

Data Science branched off my Biology line in Year 12, when an air-quality project showed me I enjoy the analysis more than the collection. Since then I have modelled algae growth for my IA and I am predicting dengue outbreaks from weather data for my Extended Essay.

Mar 2026

Air-quality data report ◌ Student-added

Maya cleaned and analysed 40,000 readings from the sensor network in Python. Her report showed a PM2.5 spike in the canteen at lunchtime, caused by the grill extractor, and facilities moved the extractor vent.



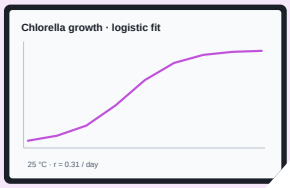
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>
> *I'm nervous about calling it a "line" when it's only a few months old. But the map shows it clearly: everything I've done since March has run along it.*

Apr 2026

Biology IA: modelling algae growth ○ Teacher-endorsed NIST-1TGTG6

For her Biology IA, Maya grew algae at five temperatures for twelve weeks, counted cells with a haemocytometer and fitted a logistic growth model in Python.

> Reflection: What was new here?
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> *Fitting a model rather than just drawing a line. The model told me something the graph did not: growth peaks earlier at higher temperatures.*

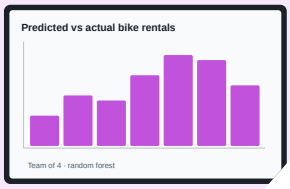


Jul 2026

Data Science summer school ◌ Student-added

A two-week online summer school run by a university data science department. Maya's team modelled bike-share demand against weather and presented to faculty.

> Reflection: Did it confirm the direction?
>
> *Yes. Two weeks of nothing but data and I wanted more, not less.*



Jan 2027

Extended Essay final submission ◌ Student-added

Planned: final Extended Essay submission in January, after a second draft and a viva with her supervisor.

Jul 2027

Target: BSc Data Science ◌ Student-added

Planned: a BSc in Data Science, ideally a course with environmental or public-health applications.

Science · Biology

Biology gave me my questions: microplastics in canals, enzyme kinetics, algae growth. Each project pushed me toward more careful measurement.

Oct 2023

Cell biology lab report ● School-verified `NIST-1QOU9Q`

First formal lab report in MYP Science. Maya prepared slides of onion epidermis and cheek cells, drew labelled diagrams at ×400 and compared plant and animal cell structures.

> Reflection: What would you do differently next time?
>
> *My drawings were accurate but I did not say how I chose which cells to draw. Next time I will explain my sampling.*



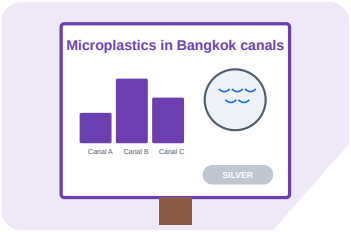
May 2024

Science fair: microplastics ◌ Student-added

For the Year 10 science fair Maya filtered water from three Bangkok canals, counted microplastic fibres under a microscope and compared the canals. The project won Silver.

Reflection: What surprised you?

Counting fibres by hand took forever. I had three jars of canal water, a cheap microscope from the science store and a tally sheet, and every session ended with sore eyes and numbers I didn't fully trust.



Halfway through, Mr Okafor asked how I knew I hadn't counted the same fibre twice. I didn't. So I started photographing each slide and marking fibres on the photo instead, which was slower but meant someone else could check me.

What surprised me most wasn't the result (canal B was much worse) but how much of science is bookkeeping. I kept thinking a program could do the counting. I didn't know how to write one yet.

Jun 2024

Reflection: why data keeps pulling me in ◌ Student-added

End-of-Year-10 reflection written for Maya's advisor. She looked back at three projects and noticed that each one turned on careful counting.

Reflection: What connected your projects this year?

Looking back at Year 10, my projects look unrelated on paper: a composting audit, a science fair, a quiz website. But they were all really about counting things carefully and finding a pattern.



The best part of the composting audit wasn't the bin we eventually got. It was the spreadsheet: six weeks of canteen waste, and the moment the chart showed Mondays were twice as bad as Fridays. I sat with it for an hour trying to work out why (it was the weekend fridge clear-out).

I want next year to have fewer projects and deeper ones. And I want to learn enough code to stop doing the boring parts by hand.

Nov 2024

Genetics unit test ● School-verified `NIST-Q27KO6`

Year 11 Biology test on Mendelian genetics, Punnett squares and pedigree analysis. Maya scored a 7, the top grade.

Reflection: How did you prepare?

I made a Punnett square for every family in a TV series. It sounds silly, but I never got a cross wrong in the test.



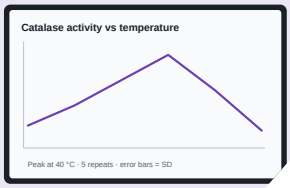
Mar 2025

Enzyme kinetics investigation ○ Teacher-endorsed `NIST-T437AV`

An independent investigation into how temperature affects catalase activity in potato, measured by oxygen volume. Maya ran five repeats at each of six temperatures and plotted means with error bars.

Reflection: What did error bars change?

They showed which differences were real. Two of my "trends" disappeared once I drew them.



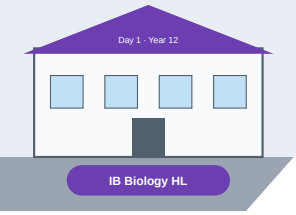
Aug 2025

Started IB Biology HL ● School-verified `NIST-1VETTI`

Maya began IB Biology at Higher Level with Mr Okafor, the start of her Diploma Programme science line.

Reflection: Why Biology at HL?

Because every project I care about starts with a living-world question: algae, mangroves, dengue.



Jan 2026

Biology IA proposal ● School-verified `NIST-1SWXEA`

Maya's proposal for her Biology internal assessment, on how water temperature affects the growth rate of Chlorella algae. It was approved with one change: add a light-intensity control.

Reflection: What did the feedback change?

I had forgotten that light varies across the lab. Now every flask sits under the same lamp.



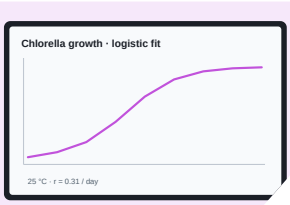
Apr 2026

Biology IA: modelling algae growth ○ Teacher-endorsed `NIST-1TGTG6`

For her Biology IA, Maya grew algae at five temperatures for twelve weeks, counted cells with a haemocytometer and fitted a logistic growth model in Python.

Reflection: What was new here?

Fitting a model rather than just drawing a line. The model told me something the graph did not: growth peaks earlier at higher temperatures.



Computing

I learned to code in Year 10 and use it as a tool: sensors, dashboards and a hackathon chatbot.

Mar 2024

Recycling quiz web app ◌ Student-added

A ten-question quiz about which bin each item goes in, built with HTML, CSS and JavaScript. It ran on the hall projector in a Year 7 assembly and 180 students answered on their phones.

Reflection: What did you learn from real users?

Half the hall got the coffee-cup question wrong, which was the point. But the buttons were too small on phones, so I fixed them that afternoon.



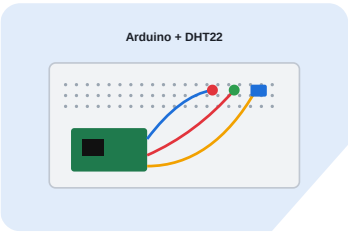
Oct 2024

Arduino workshop ● School-verified NIST-PI8DA8

A two-session maker club workshop. Maya wired a DHT22 temperature sensor to an Arduino Uno and logged a reading every ten seconds to a laptop over serial.

Reflection: What did this workshop unlock?

It was the first time code did something in the physical world. The temperature graph moved when I breathed on the sensor and I laughed out loud.



May 2025

Data visualisation course ◌ Student-added

A self-paced online course in Python data analysis. Maya completed eight graded notebooks covering pandas, cleaning, grouping and matplotlib charts.

Reflection: Why take a course on your own?

I was tired of copying formulas in spreadsheets. I used my own litter data for every exercise, which made it stick.



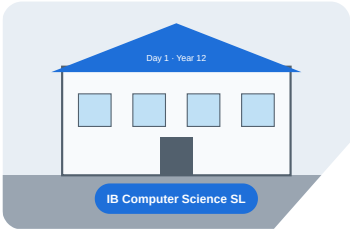
Aug 2025

Started IB Computer Science SL ● School-verified NIST-1U0VT2

Maya began IB Computer Science at Standard Level with Ms Nair, choosing it over a third science.

Reflection: Why CS over a third science?

Because I kept needing code to answer my science questions. It felt like choosing a tool I would use every day.



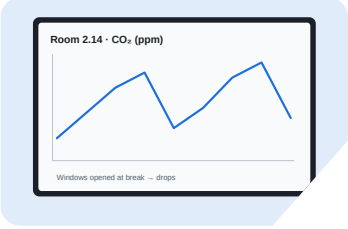
Nov 2025

CO_2 sensor prototype ◌ Student-added

A first prototype: a single CO_2 sensor on a Raspberry Pi in Maya's classroom, logging every minute. It showed CO_2 above 1,500 ppm by the end of double lessons.

Reflection: Why start with just one sensor?

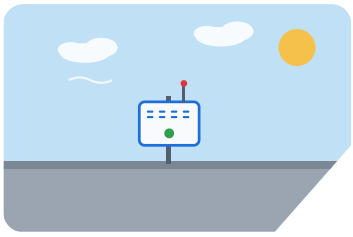
To find out what breaks. It was the power supply, twice.



Mar 2026

Air-quality sensor network ○ Teacher-endorsed `NIST-1TGBH2`

Maya built six sensor units from Raspberry Pis and PM2.5/CO_2 modules, 3D-printed weatherproof housings and mounted them around campus. She calibrated each unit against the school weather station and wrote a live web dashboard that facilities now checks daily.

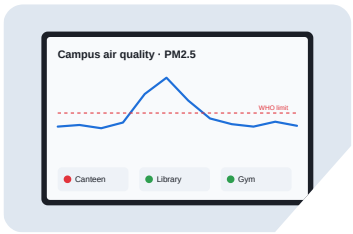


> Reflection: This project touches two lines. What joined them?
>
> *I started this as an environment project: the canteen felt smoky at lunch and I wanted proof. Building the six sensors was the fun part. The hard part was everything after.*
>
> *Two sensors drifted. One died for four days when a cleaner unplugged it, and the gap made the dashboard lie. I spent more time cleaning data than soldering: spotting the broken unit from its readings alone, deciding how to fill gaps honestly, checking every unit against the school weather station.*
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> *That's when I realised the part I enjoy most isn't the hardware or even the environment question. It's turning messy readings into something someone can trust and act on. Ms Nair said the calibration notes were the most impressive part, and I think she's right. That's why I want to study data.*

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Maya cleaned and analysed 40,000 readings from the sensor network in Python. Her report showed a PM2.5 spike in the canteen at lunchtime, caused by the grill extractor, and facilities moved the extractor vent.



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Sep 2026

Youth hackathon: 2nd place ● School-verified `NIST-1T6TUH`

A 24-hour youth hackathon on climate resilience. Maya's team of three built a LINE chatbot that sends flood alerts by district, using live rainfall data. They placed 2nd of 22 teams.



> Reflection: What was your role?
>
> *Data pipeline. I pulled the rainfall feed and decided when an alert should fire. At 4 a.m. I learned that thresholds are a design decision.*

Environment

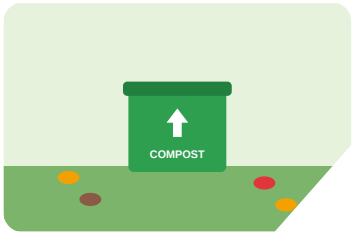
Most of my projects start with a local environmental problem I can measure.

Jan 2024

Campus composting audit ○ Teacher-endorsed NIST-1QEUO1

Over six weeks Maya and two friends weighed canteen food waste every lunchtime. They found 38 kg a week and presented a proposal for a compost bin to the facilities manager, which was approved.

> Reflection: What made the proposal work?
>
> *We had six weeks of numbers, not just an opinion. The facilities manager asked for the spreadsheet before he said yes.*



Apr 2024

Beach cleanup + litter survey ○ Teacher-endorsed NIST-1TQPWD

Maya noticed the club collected 30 bags of rubbish each month but nobody knew what was in them. She designed a tally sheet with 12 litter categories, trained 40 volunteers in ten minutes, and turned the results into a chart for the club newsletter.

> Reflection: What changed when you started counting?
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> *The cleanup stopped being just a morning out. When we saw that 40% was single-use bottles, the club started a refill-station campaign.*

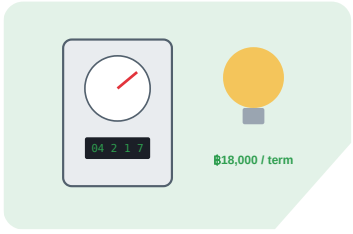


Jan 2025

Eco-committee energy audit ○ Teacher-endorsed NIST-QC76D5

With the eco-committee, Maya read every classroom electricity meter at 7:30 and 16:30 for a month. Lights and air-con left on after school were costing about ฿18,000 a term. Facilities installed timers in twelve rooms.

> Reflection: Why did this audit work?
>
> *A number in baht made people listen. "Turn the lights off" never did.*

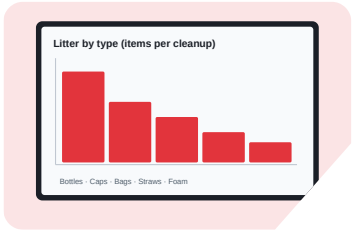


Apr 2025

Led the litter survey team ○ Teacher-endorsed NIST-1V4U7T

As survey lead, Maya ran three survey days, merged two years of tally data and wrote a two-page summary for the district environment office, which now asks the club for figures each term.

> Reflection: How did it feel to lead?
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> *Scary at first. I learned that a good survey sheet does half the leading for you.*

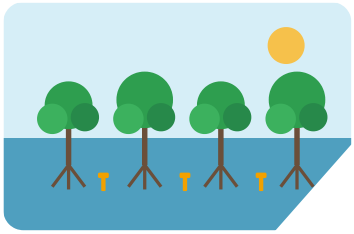


Jun 2025

Mangrove planting trip ○ Teacher-endorsed NIST-1VOTF7

A CAS trip with 30 students to Chanthaburi. Over three days they planted 600 mangrove seedlings and tagged a sample of 50 so that survival could be checked six months later.

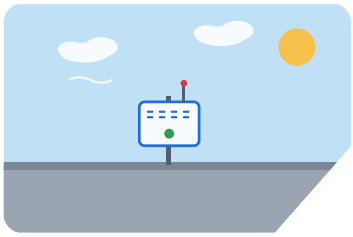
> Reflection: What will you remember?
>
> *Mud up to my knees, and the ranger telling us only half would survive. So we tagged them, to find out which half.*



Mar 2026

Air-quality sensor network ○ Teacher-endorsed NIST-1TGBH2

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Service

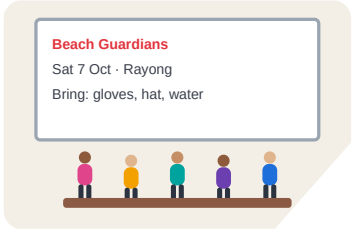
Beach Guardians and the STEM club taught me that data is only useful when someone can act on it.

Sep 2023

Joined Beach Guardians ● School-verified NIST-1QYTVF

Beach Guardians runs a cleanup on the first Saturday of every month at Mae Ramphueng beach in Rayong. Maya joined as a volunteer in Year 10 and became data lead a year later.

> Reflection: Why this club?
>
> *My friend Ploy dragged me along. I stayed because the beach looked completely different by lunchtime, and I could see what we had done.*



Apr 2024

Beach cleanup + litter survey ○ Teacher-endorsed NIST-1TQPWD

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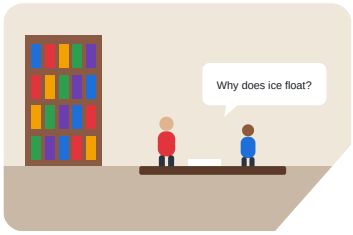
Oct 2024

Tutoring Y8 science ○ Teacher-endorsed `NIST-PS7YZ7`

Every Wednesday lunchtime for two terms, Maya tutored three Year 8 students in the library on science homework, mostly states of matter and forces.

Reflection: What does tutoring teach the tutor?

If I could not explain something with a drawing, I did not really understand it. I re-learned density properly.



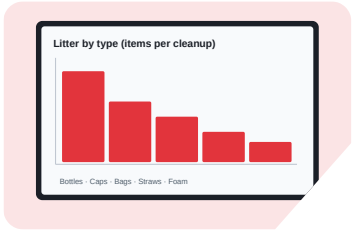
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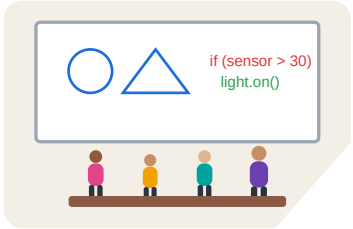
Oct 2025

CAS: STEM club for Y7 girls ○ Teacher-endorsed `NIST-1TQW7D`

Maya co-founded a Friday after-school STEM club for Year 7 girls with a friend. Over two terms, 14 regular members built light sensors, a micro:bit pet and a class weather station.

Reflection: What made the girls keep coming?

Seeing their own sensor change on the screen. Theory lost them; live data kept them.



● **School-verified** ○ **Teacher-endorsed** ◌ **Student-added**

Every item shows how it was verified. Students choose and order items but cannot edit verified facts.

NIST International School

Maya Chen · NIST Transit Portfolio

Printed 27 October 2026.

Each verified item carries a code starting NIST-. Anyone can check a code at **localhost:4961/#/v**. The school confirms what came from its systems or was endorsed by a teacher.